

Multichamber Muffler System

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Multicomponent Muffler Internal Geometry



Dimensions

Schematic Variants for Muffler Subcomponents



Part 1 —
Chamber and
Baffle



Part 2 — Fluid
domain



Part 3 —
Fiberglass

Part 4 —
Showing
perforates (aimed
at fiberglass)

Ansys Simulation

Simulated Transmission Loss (0–1000 Hz) by approximating
muffler walls as fluid at 20 deg C

 Transmission Loss at 20°C

Figure: Transmission Loss curve of the
muffler between 5 Hz and 1000 Hz at 20°C.

Simlab Simulation

Simulated Transmission Loss (0–1000 Hz) Simlab model

 Muffler TL Plot

Sidlab and Ansys File Download Center

SIDLAB Model

- **File:**
Mark3Sid.zip
- **Created with:**
SIDLAB 5.1

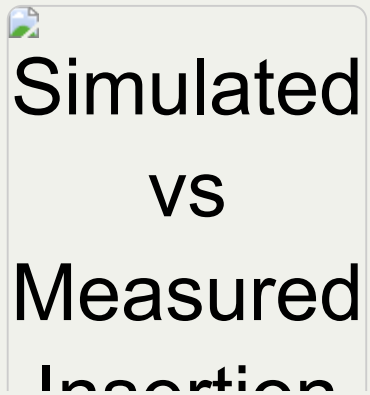
ANSYS Simulation

- **File:** Mark-I-MDF-cleaned-data.wbpz

Sidlab Components

Simulated vs Measured Insertion Loss

Measured
VS
Simulated
TL

Simulated
VS
Measured
Insertion

Insertion Loss Explanation

Insertion Loss (IL) quantifies how much sound is attenuated when a muffler is added to the system.

General formula:

$$IL = 10 \log_{10} \left(\frac{P_{\text{baseline}}}{P_{\text{muffler}}} \right)$$

References

Cited Works

1. Munjal ML. *Acoustics of Ducts and Mufflers*. 2nd ed. Wiley; 2014. ISBN: 9781118443125.
<https://doi.org/10.1002/9781118443125>
2. Dokumacı E. *Duct Acoustics: Fundamentals and Applications to*